

NATIONAL SPACE & ASTRONAUTICS CONSORTIUM (NSAC)

PROJECT DESCRIPTION

Satellite Ground Station Time Sharing

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1. Project Overview

The Satellite Ground Station Time Sharing system is a database-driven system for NSAC (National Space & Astronautics Consortium). Egyptian and African universities are building small satellites called CubeSats, and each satellite can only communicate with Earth when it passes over a ground station. NSAC owns a limited network of ground stations that all these universities must share.

Today NSAC schedules satellite passes using a spreadsheet. Because many universities want the same stations at the same times, conflicts happen every week, and solving one by manual negotiation can take an hour. Scheduling is also difficult because it depends on many factors at once: satellite orbit data, frequency licensing, station capabilities, contact requirements, weather risks, credits and operational constraints.

The new system replaces the spreadsheet with one central database that stores universities, satellites, frequency licenses, ground stations, predicted passes, contact requests, scheduled contacts, pass plans, telemetry files and commands. Using this data, the system calculates when each satellite can be contacted, schedules contacts fairly, resolves conflicts automatically, and explains its decisions. It is used by NSAC staff (station operations, frequency coordination and management), by the ground stations that execute the contacts, and by university mission operators who request contacts, download their telemetry and command their own satellites.

2. What the System Does

The system allows NSAC and the universities to:

- **Register universities and satellites** – each satellite is stored with its orbit data, radio frequencies, data rate and mission priority (science, education or technology test).
- **Check frequency licenses** – a satellite can only be scheduled if its frequency license is valid on the date of the pass.
- **Manage ground stations** – each station is stored with its location, supported frequency bands and status.
- **Predict passes** – the system downloads fresh TLE orbit data at least once a day and calculates when each satellite will pass over each station for the next 7 days.
- **Submit contact requests** – a mission operator requests contacts by giving the minimum number of passes per day, the minimum duration and any mandatory passes.
- **Schedule contacts and solve conflicts** – when two satellites want the same station at the same time, the system first tries to move one satellite to another station, then to split the pass if both operators accept, and finally decides by priority. The satellite that loses gets a higher priority in the next conflict, and a rejected request shows which satellite was given the slot.
- **Manage credits** – each university has a yearly credit budget that is reduced by the minutes of contact it uses.
- **Send pass plans to stations** – each station receives its pass plan (antenna pointing angles and commands) before the pass, and can still execute it if the link to the center fails.
- **Store telemetry** – data received from a satellite during a contact is saved as a telemetry file linked to that contact, so the university can download it at any time.
- **Log remote commands** – universities can send commands to their own satellite during their own passes only, and every command is recorded.
- **Produce reports** – contacts per satellite, station load, conflicts solved and credits used.

3. Main Data Stored in the System

The database is built around the following main entities. The table shows what each entity stores and how it is related to the other entities.

Entity	What is stored	Relationships
University	The participating university, its mission operators and its yearly credit balance.	*Owns many Satellites; submits many Contact Requests.*
Satellite	Orbit data (TLE), radio frequencies, data rate and mission priority (science, education or technology test).	*Belongs to one University; has Frequency Licenses, Passes and Contacts.*
Frequency License	The licensed frequencies of a satellite and the dates the license is valid.	*Belongs to one Satellite.*
Ground Station	Location, supported frequency bands and current status.	*Has many Passes; receives many Pass Plans.*
Pass	A predicted time when a satellite is visible over a station (start and end time), calculated 7 days ahead; may be flagged as a weather risk.	*One Satellite over one Ground Station.*
Contact Request	Minimum passes per day, minimum duration and mandatory passes; frozen 24 hours before the pass.	*Made by one University for one Satellite.*
Contact	A scheduled pass; records whether it was moved or split, how a conflict was solved, and whether it succeeded or failed.	*Uses one Pass; has one Pass Plan, many Telemetry Files and Commands.*
Pass Plan	Antenna pointing angles and the commands to send during the pass.	*Belongs to one Contact; sent to one Ground Station.*
Telemetry File	Data received during a contact (5 MB – 200 MB per pass), kept for 1 year.	*Belongs to one Contact; owned by the Satellite's University.*
Command	Target satellite ID, the command sent and its log record.	*Sent during one Contact to one Satellite.*

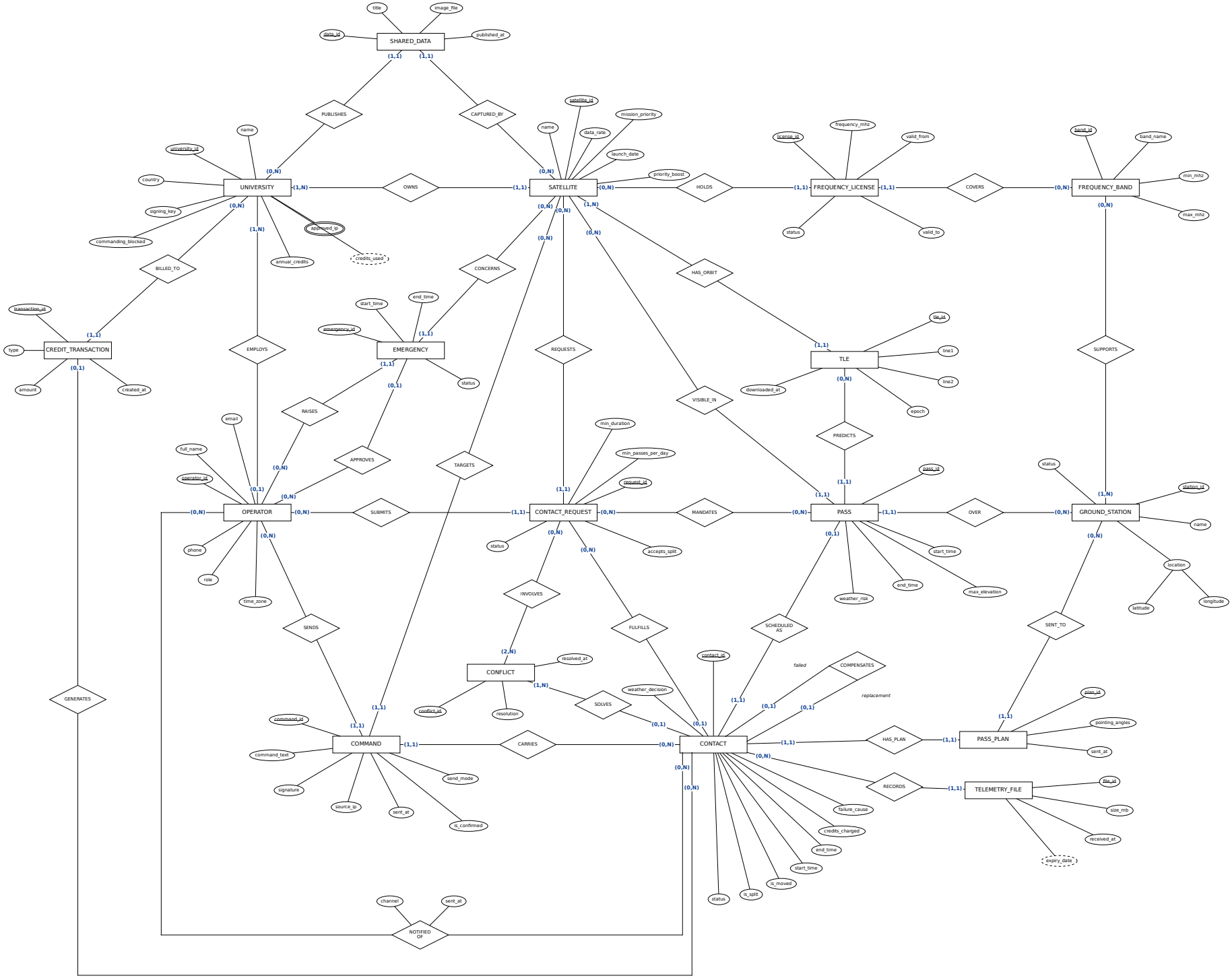
4. Main Business Rules

Category	Rules
Scheduling	▪ A satellite is scheduled only if its frequency license is valid on the date of the pass. ▪ Fresh TLE orbit data is downloaded at least once a day. ▪ A contact request is frozen 24 hours before its pass, except in an emergency. ▪ Conflicts are solved by moving → splitting (if both operators accept) → priority; the losing satellite gets a higher priority next time.
Credits	▪ Each university receives 600 credits per year; one minute of contact costs one credit. ▪ A new science mission receives 50 free passes during the first 30 days after launch. ▪ Passes lost because of a station failure are compensated within the following 7 days.
Emergency & Weather	▪ Emergency mode gives a satellite in trouble all available passes for 24 hours. ▪ Passes are flagged as a weather risk when dust storms are forecast.
Stations & Users	▪ A station receives its pass plan at least 30 minutes before the pass. ▪ Operators are notified by email or SMS 15 minutes before each of their passes. ▪ All times are shown in the user's local time zone.
Telemetry	▪ Telemetry belongs to the satellite owner and is kept by the center for 1 year.
Commanding Security	▪ Commands are allowed only during the university's own passes and every command is logged. ▪ A university can never send commands to another university's satellite.

SECTION 02 / DATABASE DESIGN

ERD

Entity-Relationship Diagram


SHARED_DATA
title
image_file
data_id
published_at
PUBLISHES
CAPTURED_BY
UNIVERSITY
university_id
name
country
signing_key
commanding_blocked
annual_credits
credits_used
approved_ip
BILLED_TO
OWNS
SATELLITE
satellite_id
name
mission_priority
data_rate
launch_date
priority_boost
HOLDS
FREQUENCY_LICENSE
license_id
frequency_mhz
valid_from
status
valid_to
COVERS
FREQUENCY_BAND
band_id
band_name
min_mhz
max_mhz
CONCERNS
HAS_ORBIT
TLE
tle_id
line1
line2
epoch
downloaded_at
VISIBLE_IN
PREDICTS
SUPPORTS
CREDIT_TRANSACTION
transaction_id
type
amount
created_at
EMPLOYS
EMERGENCY
emergency_id
start_time
end_time
status
REQUESTS
RAISES
APPROVES
TARGETS
OPERATOR
operator_id
full_name
email
phone
role
time_zone
SUBMITS
CONTACT_REQUEST
min_duration
min_passes_per_day
request_id
status
accepts_split
MANDATES
PASS
pass_id
start_time
end_time
max_elevation
weather_risk
OVER
GROUND_STATION
station_id
status
name
location
latitude
longitude
SENT_TO
SCHEDULED AS
INVOLVES
FULFILLS
SENDS
CONFLICT
conflict_id
resolved_at
resolution
SOLVES
weather_decision
GENERATES
COMMAND
command_id
command_text
signature
source_ip
sent_at
send_mode
is_confirmed
CARRIES
CONTACT
contact_id
failed
COMPENSATES
replacement
failure_cause
credits_charged
end_time
start_time
is_moved
is_split
status
HAS_PLAN
PASS_PLAN
plan_id
pointing_angles
sent_at
RECORDS
TELEMETRY_FILE
file_id
size_mb
received_at
expiry_date
NOTIFIED OF
channel
sent_at
(0,N)
(1,1)
(1,N)
(0,1)
(2,N)